

Random Password Generator

Project 3 - DecodeLabs Python Internship

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Objective

To develop a Python program that generates secure random passwords based on user-defined length using Python libraries and string manipulation techniques.

Algorithm

1. Import required modules.
2. Take password length from the user.
3. Create a character set containing letters, digits and symbols.
4. Generate random characters.
5. Join characters into a password.
6. Display the generated password.

Python Code

```
import secrets
import string

length = int(input("Enter password length: "))
characters = string.ascii_letters + string.digits + string.punctuation
password = ''.join(secrets.choice(characters) for _ in range(length))
print("Generated Secure Password:", password)
```

Features

- User-defined password length
- Secure password generation
- Uses letters, numbers and symbols
- Easy to use

Sample Output

Enter password length: 10
Generated Secure Password: A@7m#P2xL\$

Conclusion

This project successfully generates secure random passwords using Python and demonstrates practical usage of modules, string manipulation and secure coding practices.